ENGINEERING ECONOMY

LECTURE 9

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Rate of Return Analysis: Multiple Alternatives

- **Why Incremental Analysis Is Necessary?**
- When two or more mutually exclusive alternatives are evaluated, engineering economy can identify the one alternative that is the best economically. As we have learned, the PW and AW techniques can be used to do so, and are the recommended methods. Now the procedure using ROR to identify the best is presented.

- Let’s assume that a company uses a MARR of 16% per year, that the company has $90,000 available for investment, and that two alternatives (A and B) are being evaluated.
- Alternative A requires an investment of $50,000 and has an internal rate of return i_A^* of 35% per year.
- Alternative B requires $85,000 and has an i_B^* of 29% per year.
- Intuitively we may conclude that the better alternative is the one that has the larger return, A in this case.
- However, this is not necessarily so. While A has the higher projected return, its initial investment ($50,000) is much less than the total money available ($90,000).

- What happens to the investment capital that is left over? It is generally assumed that excess funds will be invested at the company's MARR, as we learned in previous chapters.
- Using this assumption, it is possible to determine the consequences of the two alternative investments.
- If alternative A is selected, $50,000 will return 35% per year.
- The $40,000 left over will be invested at the MARR of 16% per year. The rate of return on the total capital available, then, will be the weighted average.

- Thus, if alternative A is selected,

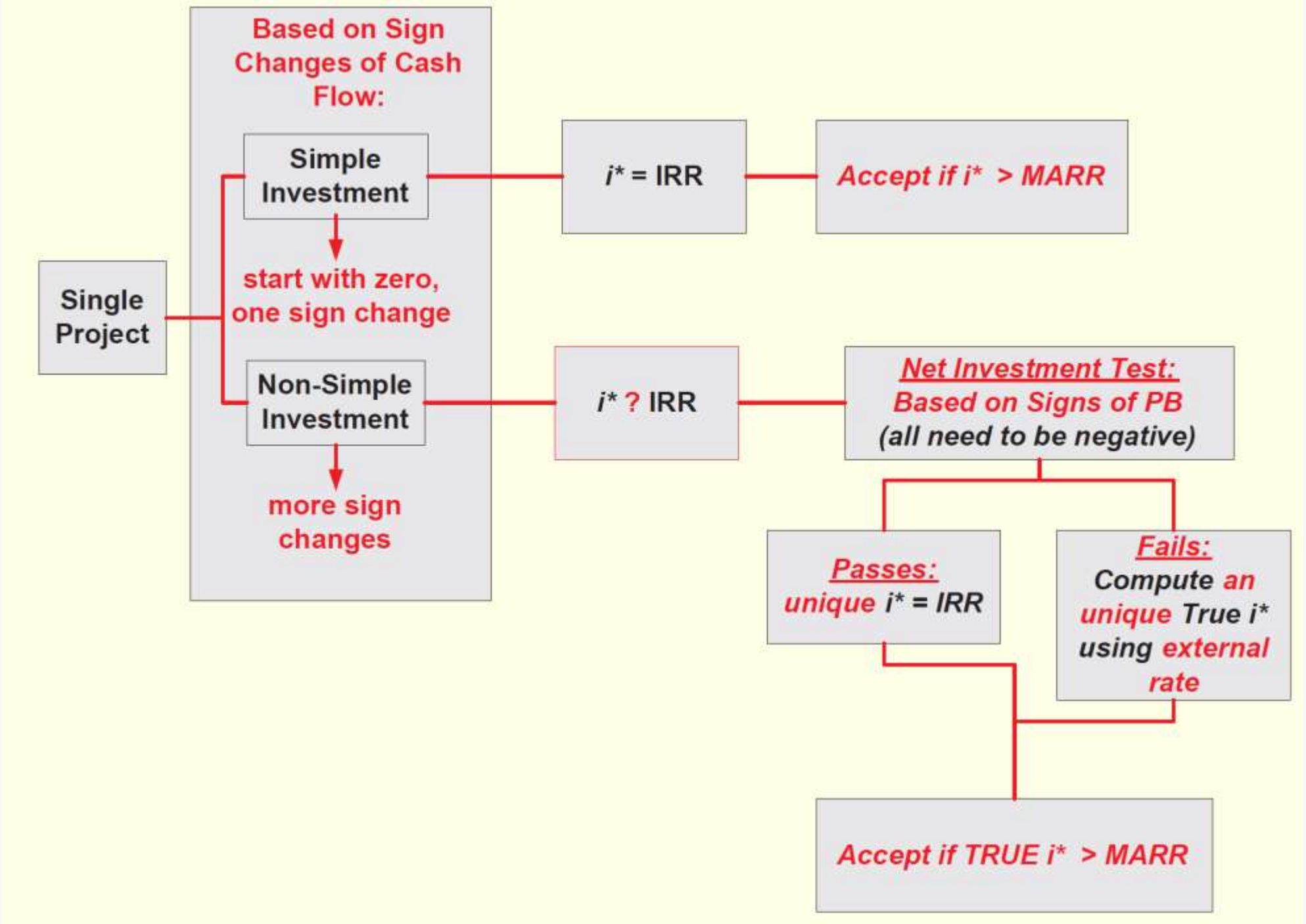
$$\text{Overall ROR}_A = \frac{50{,}000(0.35) + 40{,}000(0.16)}{90{,}000} = 26.6\%$$

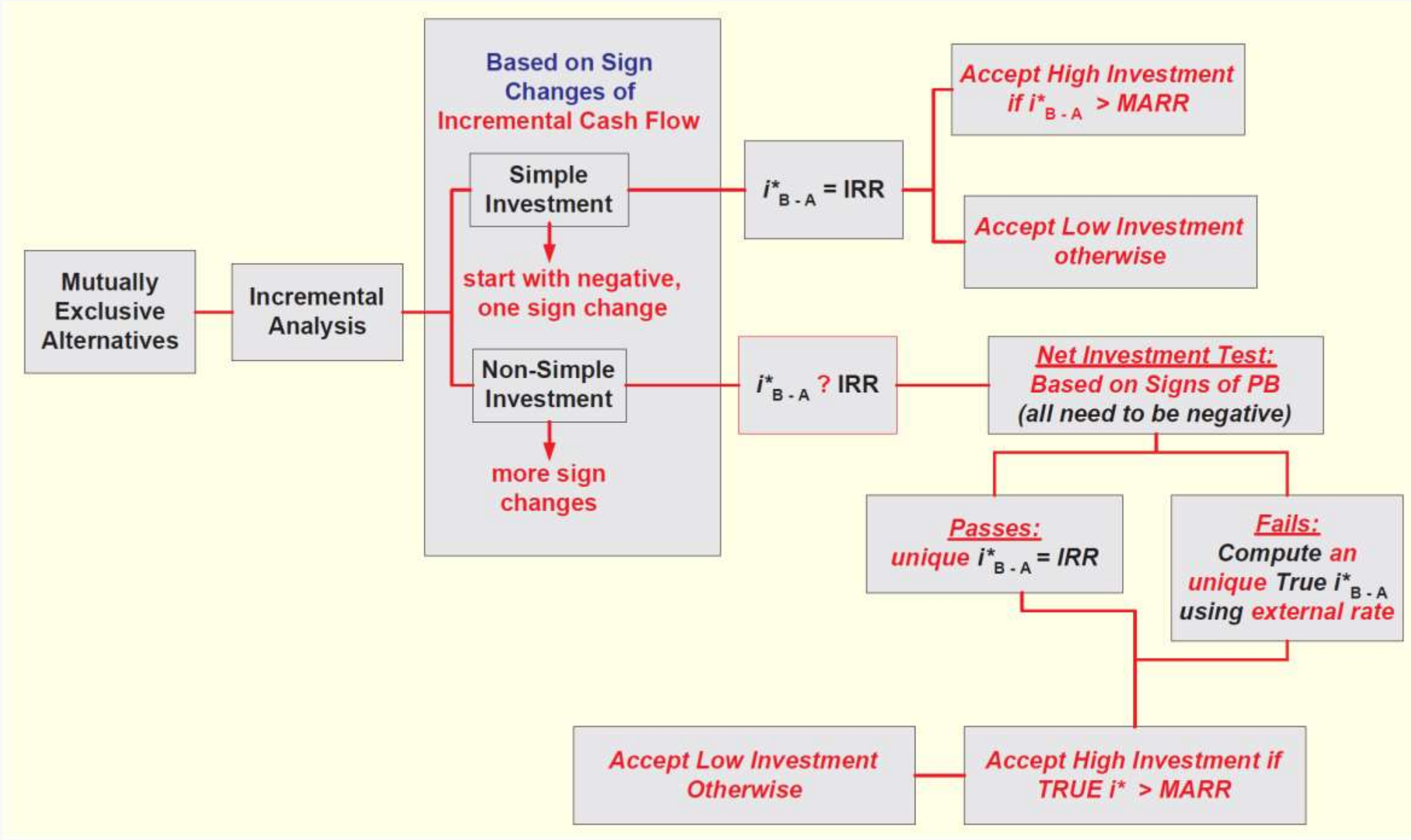
- If alternative B is selected, $85,000 will be invested at 29% per year, and the remaining $5000 will earn 16% per year. Now the weighted average is

$$\text{Overall ROR}_B = \frac{85{,}000(0.29) + 5000(0.16)}{90{,}000} = 28.3\%$$

- These calculations show that even though the *i** for alternative A is higher, alternative B presents the better overall ROR for the $90,000. If either a PW or AW comparison is conducted using the ARR of 16% per year as *i*, alternative B will be chosen.
- This simple example illustrates a major fact about the rate of return method for ranking and comparing alternatives:
- Under some circumstances, project ROR values do not provide the same ranking of alternatives
- as do PW and AW analyses. This situation does not occur if we conduct an **incremental ROR analysis**

- When **independent projects** are evaluated, no incremental analysis is necessary between projects.
- Each project is evaluated separately from others, and more than one can be selected.
- Therefore, the only comparison is with the do-nothing alternative for each project.
- The Project ROR can be used to accept or reject each one.


Single Project
Based on Sign Changes of Cash Flow:
Simple Investment
start with zero, one sign change
Non-Simple Investment
more sign changes
i* = IRR
Accept if i* > MARR
i* ? IRR
Net Investment Test:
Based on Signs of PB
(all need to be negative)
Passes:
unique i* = IRR
Fails:
Compute an unique True i* using external rate
Accept if TRUE i* > MARR


Mutually Exclusive Alternatives
Incremental Analysis
Based on Sign Changes of Incremental Cash Flow
Simple Investment
start with negative, one sign change
Non-Simple Investment
more sign changes
i*B - A = IRR
Accept High Investment if i*B - A > MARR
Accept Low Investment otherwise
i*B - A ? IRR
Net Investment Test: Based on Signs of PB (all need to be negative)
Passes: unique i*B - A = IRR
Fails: Compute an unique True i*B - A using external rate
Accept High Investment if TRUE i* > MARR
Accept Low Investment Otherwise

Calculation of Incremental Cash Flows for ROR Analysis

- To conduct an incremental ROR analysis, it is necessary to calculate the **incremental cash flow** series over the lives of the alternatives. Based upon the equivalence relations (PW and AW), ROR evaluation makes the equal-service assumption.
- The incremental ROR method requires that the equal-service requirement be met.
- Therefore, the **LCM (least common multiple) of lives for each pairwise comparison** must be used.
- All the assumptions of equal service present for PW analysis are necessary for the incremental ROR method.

- Equal-life alternatives have *n* years of incremental cash flows, while unequal-life alternatives require the LCM of lives for analysis. At the end of each life cycle, the salvage value and initial investment for the next cycle must be included for the LCM case.
- When a **study period** is established, only this number of years is used for the evaluation.
- All incremental cash flows outside the period are neglected.
- As we learned earlier, using a study period, especially one shorter than the life of either alternative, can change the economic decision from that rendered when the full lives are considered.

- Only for the purpose of simplification, use the convention that between two alternatives, the one with the *larger initial investment* will be regarded as *alternative B*. Then, for each year

$$\textbf{Incremental cash flow} = \textbf{cash flow}_B - \textbf{cash flow}_A$$

- The initial investment and annual cash flows for each alternative (excluding the salvage value) are one of the types:
- *Revenue alternative*, where there are both negative and positive cash flows
- *Cost alternative,* where all cash flow estimates are negative

EXAMPLE

- Liberty Pharmacy must improve its point-of-sale (POS) speed for customers to remain competitive with large chains such as Walgreen's and CVS. Liberty owners plan to replace all scanner and cash register equipment in their six locations throughout Austin, Texas. With the decreasing costs and improved technology, two options are available. New systems will cost a total of $21,000; however, slightly used equipment is available for $15,000. Because the new systems are more sophisticated, their annual operating cost is expected to be $7000 per year, while the used systems are expected to require $8200 per year. The expected life is 10 years for both systems with a 5% salvage value. Tabulate the incremental cash flow series.

Cash Flow Tabulation for Example 8.1

Year	Cash Flow		Incremental Cash Flow (New – Used)
	Used Press	New Press	
0	$–15,000	$–21,000	$–6,000
1–10	–8,200	–7,000	+1,200
10	+750	+1,050	+300

- The salvage values in year 10 are separated from ordinary cash flow for clarity. When disbursements are the same for a number of consecutive years, *for hand solution only*, it saves time to make a single cash flow listing, as is done for years 1 to 10.

EXAMPLE

- A sole-source vendor can supply a new industrial park with large transformers suitable for underground utilities and vault-type installation. Type A has an initial cost of $70,000 and a life of 8 years. Type B has an initial cost of $95,000 and a life expectancy of 12 years. The annual operating cost for type A is expected to be $9000, while the AOC for type B is expected to be $7000. If the salvage values are $5000 and $10,000 for type A and type B, respectively, tabulate the incremental cash flow using their LCM for hand and spreadsheet solutions.

- The LCM of 8 and 12 is 24 years. In the incremental cash flow tabulation for 24 years, note that the reinvestment and salvage values are shown in years 8 and 16 for type A and in year 12 for type B.

Incremental Cash Flow Tabulation, Example 8.2

Year	Cash Flow		Incremental Cash Flow (B – A)
	Type A	Type B	
0	$ −70,000	$ −95,000	$−25,000
1–7	−9,000	−7,000	+2,000
8	−70,000 −9,000 +5,000	−7,000	+67,000
9–11	−9,000	−7,000	+2,000
12	−9,000	−95,000 −7,000 +10,000	−83,000
13–15	−9,000	−7,000	+2,000
16	−70,000 −9,000 +5,000	−7,000	+67,000
17–23	−9,000	−7,000	+2,000
24	−9,000 +5,000	−7,000 +10,000	+7,000
	$−411,000	$−338,000	$+73,000

Interpretation of Rate of Return on the Extra Investment

Cash Flow Tabulation for Example 8.1

Year	Cash Flow		Incremental Cash Flow (New – Used)
	Used Press	New Press	
0	$–15,000	$–21,000	$–6,000
1–10	–8,200	–7,000	+1,200
10	+750	+1,050	+300

- Table reflects the *extra investment* or *cost* required if the alternative with the larger first cost is selected. This is important in an incremental ROR analysis in order to determine the ROR earned on the extra funds expended for the largerinvestment alternative. If the incremental cash flows of the larger investment don't justify it, we must select the cheaper one. If there is no cheaper, justified alternative, the DN alternative is the best economic choice.

Cash Flow Tabulation for Example 8.1

Year	Cash Flow		Incremental Cash Flow (New – Used)
	Used Press	New Press	
0	$–15,000	$–21,000	$–6,000
1–10	–8,200	–7,000	+1,200
10	+750	+1,050	+300

- The new systems require an extra investment of $6000. If the new systems are purchased, there will be a “savings” of $1200 per year for 10 years, plus an extra $300 in year 10.
- The decision to buy the used or new systems can be made on the basis of the profitability of investing the extra $6000 in the new systems. If the equivalent worth of the savings is greater than the equivalent worth of the extra investment at the MARR, the extra investment should be made (i.e., the larger first-cost proposal should be accepted).
- On the other hand, if the extra investment is not justified by the savings, select the lower-investment proposal.

- If the rate of return available through the incremental cash flow equals or exceeds the MARR, the alternative associated with the extra investment should be selected.
- Not only must the return on the extra investment meet or exceed the MARR, the return on the investment that is common to both alternatives must meet or exceed the MARR. Accordingly, prior to performing an incremental ROR analysis, it is advisable to determine the internal rate of return i^* for each alternative. This can be done only for *revenue alternatives* because cost alternatives have only cost (negative) cash flows and no i^* can be determined. The guideline is as follows:
- **For multiple revenue alternatives,** calculate the internal rate of return i^* for each alternative, and eliminate all alternatives that have an $i^* < \text{MARR}$. Compare the remaining alternatives incrementally.

- **MARR and IRR: the basic decision rule**
- The **Minimum Attractive Rate of Return (MARR)** is the minimum return that a project must earn to be acceptable. It represents the decision maker's opportunity cost or required rate of return.
- The **internal rate of return (IRR)**, also called the **rate of return (ROR)**, is the interest rate that makes the net present worth (NPW) of a project equal to zero.
- **Basic rule for a single project:**
  - If the IRR of a project is **greater than or equal to the MARR**, the project is economically acceptable.
  - If the IRR is **less than the MARR**, the project should be rejected.

- **Screening alternatives using IRR**
- Suppose the MARR is **15%** and two alternatives have IRR values of **12%** and **21%**.
- The alternative with **12%** can be eliminated immediately because it does not meet the MARR.
- With only two alternatives, it is obvious that the **21% alternative** should be selected.
- If **both alternatives** have IRR values **below the MARR**, then **neither project is justified**, and the best economic choice is the **do-nothing alternative**.
- When **three or more alternatives** are being evaluated, it is often useful (though not required) to compute the IRR of each alternative as a **preliminary screening step**. Any alternative with IRR < MARR can be removed from further analysis.
- This screening step is especially convenient when using a spreadsheet, since the **IRR function** can quickly identify unacceptable alternatives.

- **Incremental IRR for mutually exclusive alternatives**
- When alternatives are **mutually exclusive** (only one can be selected), comparing individual IRR values can be misleading. In this case, the correct approach is to compute the **incremental IRR**, denoted i^*_{B-A} , based on the **incremental cash flow** between two alternatives B and A.
- The incremental IRR measures the rate of return on the **additional investment** required to move from alternative A to alternative B.

- **Relationship between individual IRRs and incremental IRR**
- Assume that **alternative B requires a larger initial investment than alternative A.**
- If

$$IRR_B < IRR_A$$

- then the incremental IRR satisfies

$$i^*_{B-A} < IRR_B$$

- **Interpretation:**
  A project with a lower IRR that also requires more investment will produce an even lower return on the additional investment.

- If

$$IRR_B > IRR_A$$

- Then

$$i^*_{B-A} > IRR_B$$

- **Interpretation:**
  When the higher-investment project also has a higher IRR, the return on the additional investment is even greater.
- This result follows from the weighted-average nature of rate-of-return calculations.

Independent projects

- For **independent projects**, there is **no comparison of additional investment**, because selecting one project does not prevent selecting another.
- If there is **no budget limitation**, the decision rule is simple:
- Accept all projects with IRR ≥ MARR.

Example:

Let MARR = 10%, and suppose three independent projects have the following IRRs:

- Project A: 12%
- Project B: 9%
- Project C: 23%
- Projects **A and C** are accepted because their IRRs exceed the MARR. Project **B** is rejected because its IRR is below the MARR.

- **IRR is a screening tool:** projects with IRR below MARR are rejected. **Incremental IRR is a decision tool:** it determines which mutually exclusive alternative should be selected.

DECISION RULES:

- **IRR is mainly a screening tool**
- Quickly eliminates projects that do not meet the MARR
- Can mislead when:
  - *projects are mutually exclusive,*
  - *project sizes differ,*
  - *cash flows are non-conventional.*
- **IRR is mainly a screening tool**
- Quickly eliminates projects that do not meet the MARR
- Can mislead when:
  - *projects are mutually exclusive,*
  - *project sizes differ,*
  - *cash flows are non-conventional.*

- **PW and AW are decision tools**
  - Always give a unique and correct ranking
  - Fully consistent with economic theory
  - Directly measure value creation

- **When to prefer AW**
- Projects have **different lives**
- Long-term planning (capacity, infrastructure, replacement analysis)
- **If projects are mutually exclusive → use PW or AW (or incremental IRR).**
  If projects are independent → IRR ≥ MARR is sufficient.

Rate of Return Evaluation Using PW: Incremental and Breakeven (Two Alternatives)

Complete Procedure for Incremental ROR using *PW*

1. Eliminate all alternatives that have an $i^* < MARR$
2. Order investments by initial investment or cost, starting with the smaller one, called A.
3. Develop the cash flow and incremental cash flow series using the LCM of years
4. Draw an incremental cash flow diagram
5. By checking the number of sign changes in the incremental cash flow series, decide if multiple rates of return may be present
6. Set up the *PW* or *AW* equation for incremental cash flows and determine a single Δi^*_{B-A}.
7. Select by the following rule:
   - If $\Delta i^*_{B-A} < MARR$, select alternative A.
   - If $\Delta i^*_{B-A} \geq MARR$, select alternative B.

EXAMPLE

A bank uses a MARR of 30%. Two alternative systems for a new online banking are under consideration. For each system, start-up, annual net income, and salvage value estimates are given:

	System A	System B
Initial Investment, $	$-12,000	$-18,000
Net Income per year, $ per year	5,000	7,000
Salvage Value, $	2,500	3,000
Estimated Life, years	8	8

Perform the Incremental ROR analysis.

$$P/A = \left[\frac{(1+i)^n - 1}{i(1+i)^n}\right]$$

$$PW_A = 0 \quad = \quad -12,000 + 5,000(P/A, i\%, 8) + 2,500(P/F, i\%, 8)$$

Initial i_A^* is found by ignoring time value of money:

$$\begin{aligned} 12,000 &= 42,500(P/F, i\%, 8) \\ (P/F, i\%, 8) &= 0.2823 \rightarrow [16\%, 18\%] \end{aligned}$$

i%	18	28	38	40
PW_A	9053	3726	347.6	-177.6

$$i_A^* = 38 + \frac{0 - 347.6}{-177.6 - 347.6}(2) = 39.32\%$$

$i_A^* > MARR = 30\%$, include in the analysis

$$PW_B = 0 \quad = \quad -18,000 + 7,000(P/A, i\%, 8) + 3,000(P/F, i\%, 8)$$

Initial i_B^* is found by ignoring time value of money:

$$\begin{aligned} 18,000 &= 59,000(P/F, i\%, 8) \\ (P/F, i\%, 8) &= 0.3051 \rightarrow 16\%, \end{aligned}$$

i%	16	26	36	37
PW_B	13320	5157	39	-363.8

$$i_B^* = 36 + \frac{0 - 39}{-39 - 363.8}(1) = 36.1\%$$

$i_B^* > MARR = 30\%$, include in the analysis

Cash Flows: Year	Alternative A	Alternative B	Incremental B - A
0	$-12,000	$-18,000	$-6,000
1-8	5,000	7,000	2,000
8	2,500	3,000	500

$$PW_\Delta = 0 \quad = \quad -6,000 + 2,000(P/A, i\%, 8) + 500(P/F, i\%, 8)$$

Initial Δi^*_{B-A} is found by ignoring time value of money:

$$6,000 \quad = \quad 16,500(P/F, i\%, 8)$$

$$(P/F, i\%, 8) \quad = \quad 0.3636 \rightarrow [12\%, 14\%]$$

i%	14	24	26	28	30
PW_Δ	3453	931.9	560	221	-89

$$\Delta i^*_{B-A} \quad = \quad 28 + \frac{0 - 221}{-89 - 221}(2) = 29.4\%$$

$\Delta i^*_{B-A} < MARR = 30\%$, Alternative A is selected.

EXAMPLE

- As Ford Motor Company of Europe retools an old assembly plant in the United Kingdom to produce a fuel-efficient economy model automobile, Ford and its suppliers are seeking additional sources for light, long-life transmissions. Automatic transmission component manufacturers use highly finished dies for precision forming of internal gears and other moving parts. Two international vendors make the required dies. Use the per unit estimates below and a MARR of 12% per year to select the more economical vendor bid. Show both hand and spreadsheet solutions.

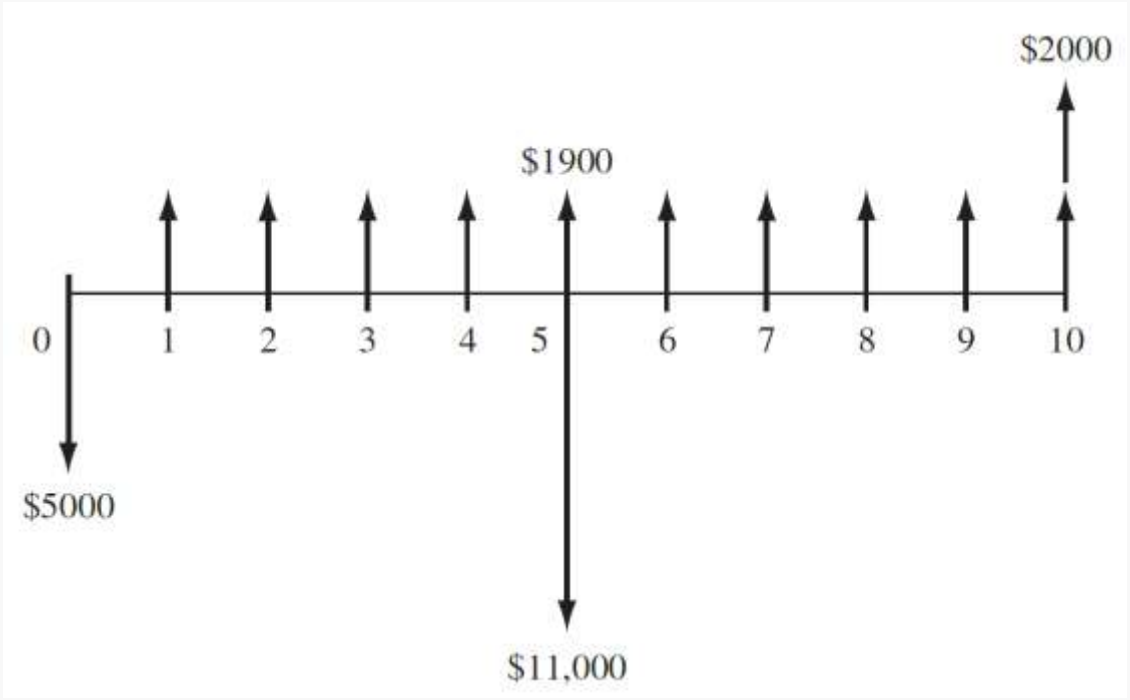
	A	B
Initial cost, $	−8,000	−13,000
Annual costs, $ per year	−3,500	−1,600
Salvage value, $	0	2,000
Life, years	10	5

These are cost alternatives, since all cash flows are costs. Use the procedure described above to determine Δi^*_{B-A}.

1. Alternatives A and B are correctly ordered with the higher first-cost alternative in column 2 of Table
2. The cash flows for the LCM of 10 years are tabulated.
3. The incremental cash flow diagram is shown in Figure
4. There are three sign changes in the incremental cash flow series, indicating as many as three roots. There are also three sign changes in the cumulative incremental series, which starts negatively at $S_0 = \$-5000$ and continues to $S_{10} = \$+5000$, indicating that more than one positive root may exist.

Incremental Cash Flow Tabulation, Example 8.3

Year	Cash Flow A (1)	Cash Flow B (2)	Incremental Cash Flow (3) = (2) − (1)
0	$ −8,000	$−13,000	$ −5,000
1–5	−3,500	−1,600	+1,900
5	—	+2,000 −13,000	−11,000
6–10	−3,500	−1,600	+1,900
10	—	+2,000	+2,000
	$−43,000	$−38,000	$ +5,000



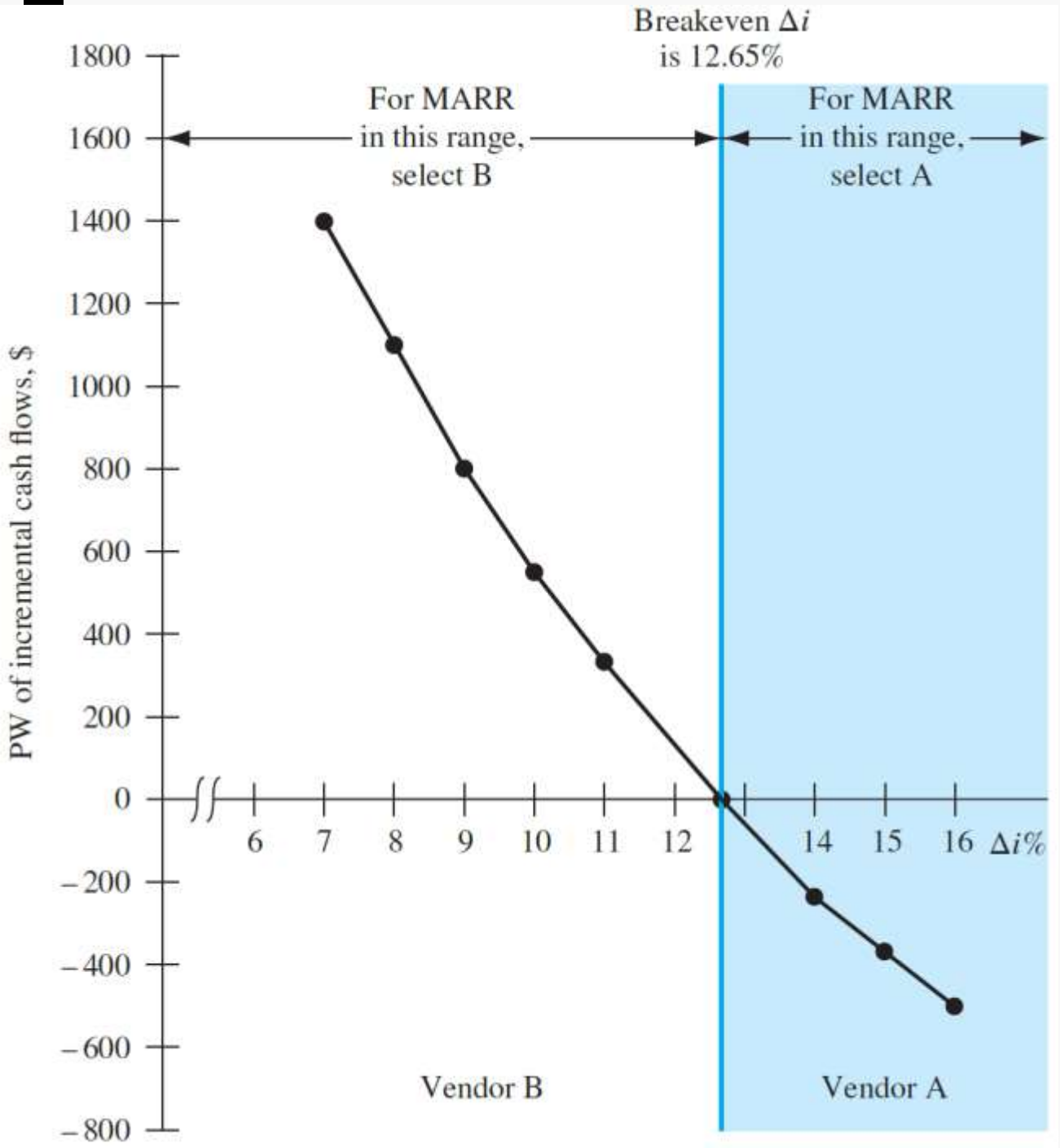
5. The rate of return equation based on the PW of incremental cash flows is

$$0 = -5000 + 1900(P/A,\Delta i^*,10) - 11{,}000(P/F,\Delta i^*,5) + 2000(P/F,\Delta i^*,10)$$

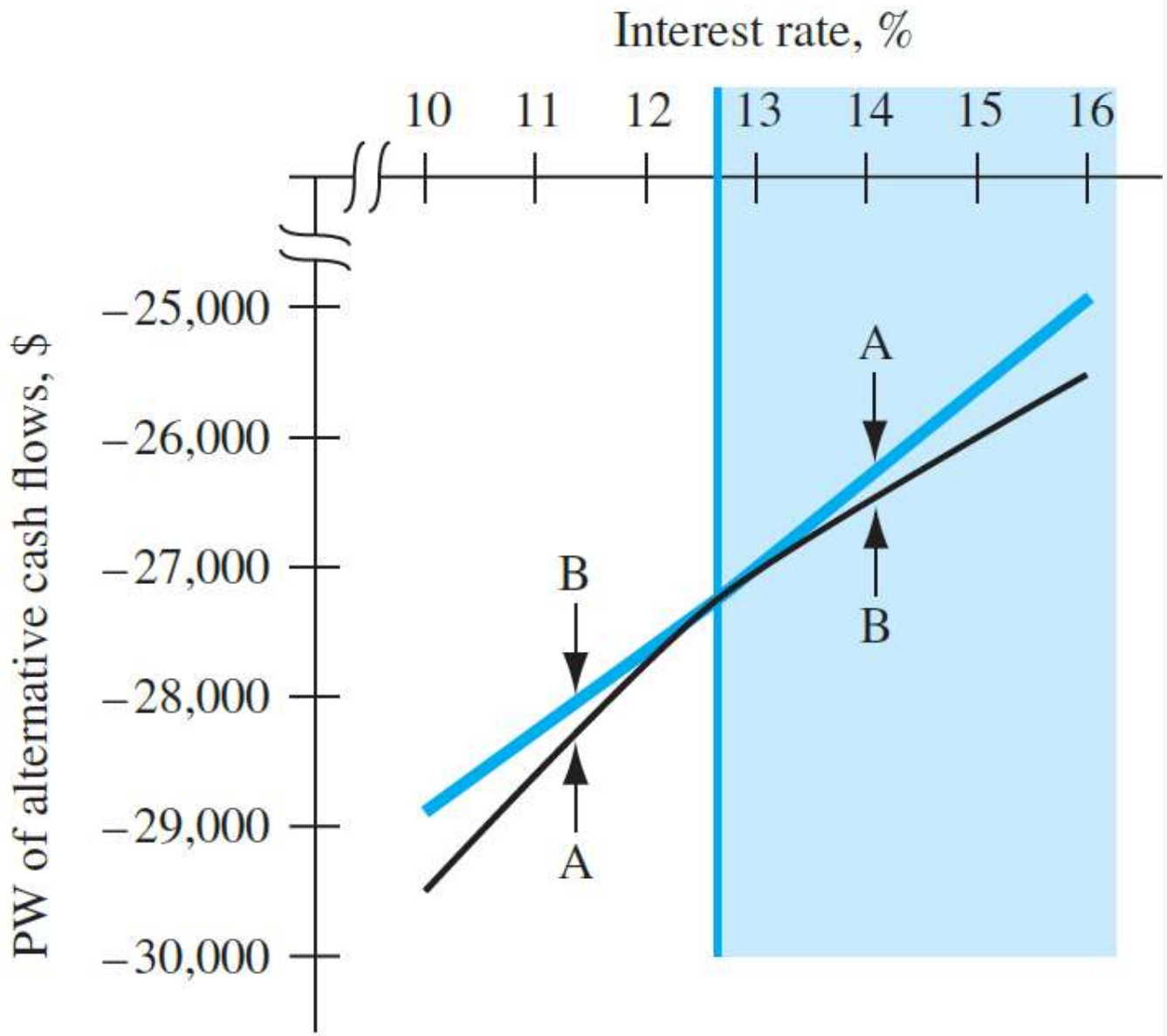
In order to resolve any multiple-root problem, we can assume that the investment rate i_i in the ROIC technique will equal the Δi^* found by trial and error. Solution of Equation for the first root discovered results in Δi^* between 12% and 15%. By interpolation $\Delta i^* = 12.65\%$.

6. Since the rate of return of 12.65% on the extra investment is greater than the 12% MARR, the higher-cost vendor B is selected.

- The **breakeven rate of return** is the incremental i^* value, Δi^*, at which the PW (or AW) value of the *incremental cash flows* is exactly zero. This is the same as stating that the breakeven ROR is the i value, i^*, at which the PW (or AW) values of two alternatives' actual cash flows are exactly equal to each other.
- If the incremental cash flow ROR (Δi^*) is greater than the MARR, the larger-investment alternative is selected.
- For example, if the PW versus Δi graph for the incremental cash flows is plotted for various interest rates, the Δi^* breakeven can be seen.



- The Δi^* breakeven is at 12.65%.
- The conclusions are that
  - For MARR < 12.65%, the extra investment for B is justified.
  - For MARR > 12.65%, the opposite is true—the extra investment in B should not be made, and vendor A is selected.
  - If MARR is exactly 12.65%, the alternatives are equally attractive.



- Figure is a breakeven graph of PW versus *i* for the cash flows (not incremental) of each alternative in the Example. It provides the same results.
- Since all net cash flows are negative (cost alternatives), the PW values are negative. Now, the same conclusions are reached using the following logic:

• If MARR < 12.65%, select B since its PW of cost cash flows is smaller (numerically larger).

• If MARR > 12.65%, select A since its PW of costs is smaller.

• If MARR is exactly 12.65%, either alternative is equally attractive.

EXAMPLE

- New filtration systems for commercial airliners are available that use an electric field to remove up to 99.9% of infectious diseases and pollutants from aircraft air. This is vitally important, as many of the flu germs, viruses, and other contagious diseases are transmitted through the systems that recirculate aircraft air many times per hour. Investments in the new filtration equipment can cost from $100,000 to $150,000 per aircraft, but savings in fuel, customer complaints, legal actions, etc., can also be sizable. Use the estimates below (in $100 units) from two suppliers provided to an international carrier to do the following, using a spreadsheet and an MARR of 15% per year.

• Plot two graphs: PW versus *i* values for both alternatives' cash flows and PW versus Δi values for incremental cash flows.

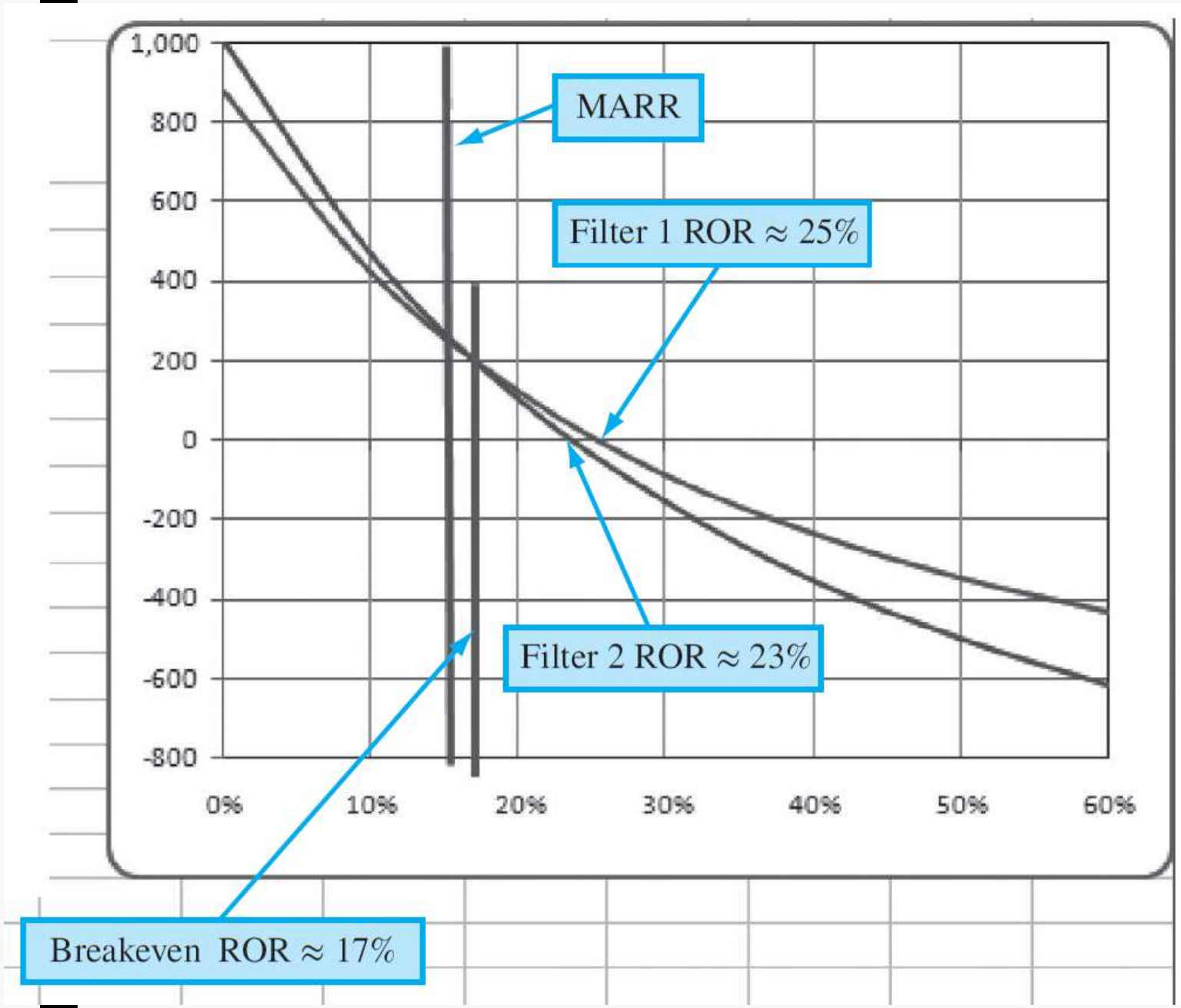
• Estimate the breakeven ROR values from both graphs, and use this estimate to select one alternative.

	Air Cleanser (Filter 1)	Purely Heaven (Filter 2)
Initial cost per aircraft, $	−1000	−1500
Estimated savings, $ per year	375	700 in year 1, decreasing by 100 per year thereafter
Estimated life, years	5	5

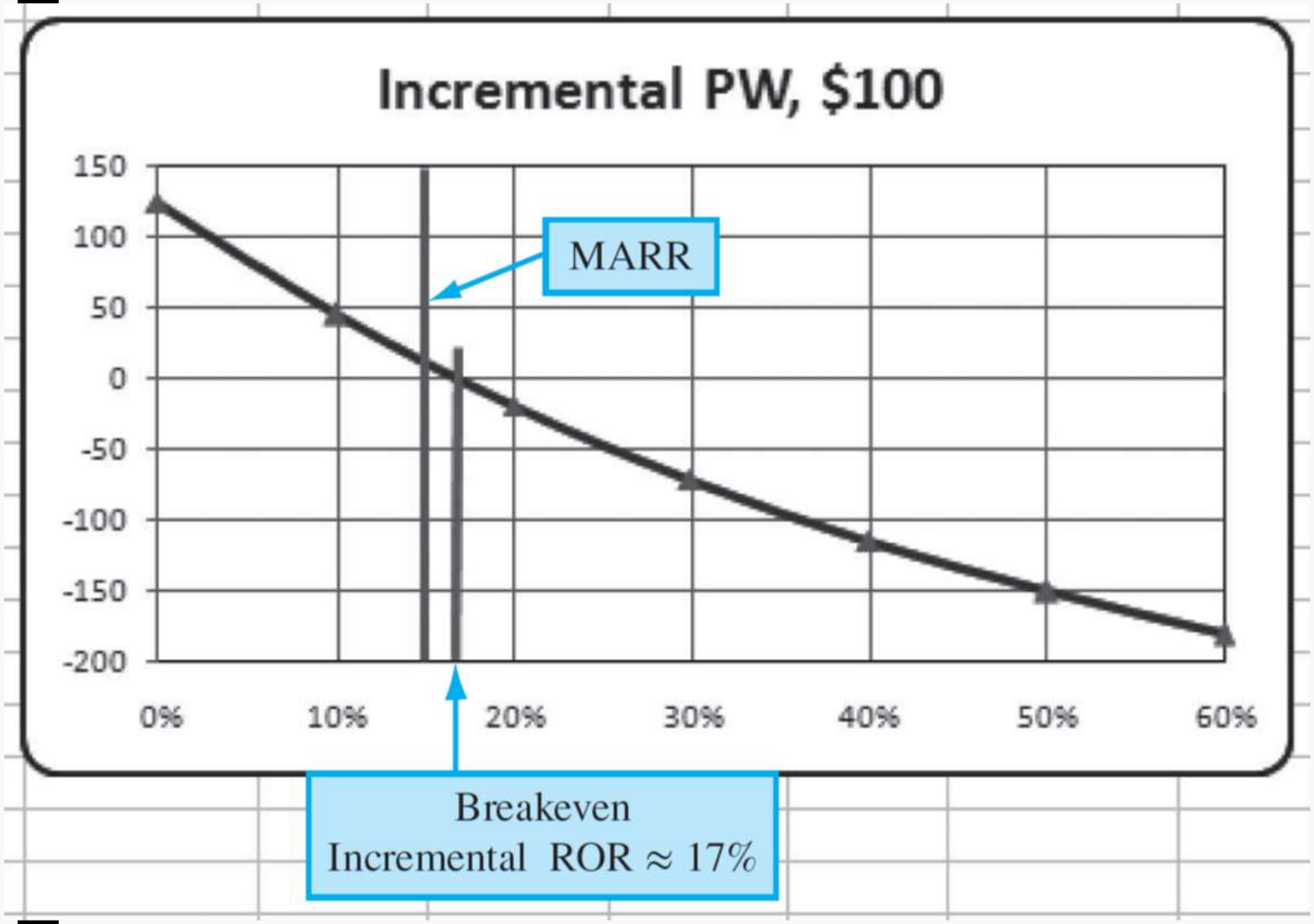
	A	B	C	D	E	F	G	H
1		Filter 1		Filter 1	Filter 2			
2	Year	Cash flow, $100	Rate, i%	PW, $ 100	Cash flow, $100	PW, $100	Incremental cash flow, $100	Incremental PW, $100
3	0	-1,000	0%	875	-1,500	1,000	-500	125
4	1	375	10%	422	700	467	325	46
5	2	375	20%	121	600	103	225	-19
6	3	375	30%	-87	500	-158	125	-71
7	4	375	40%	-237	400	-352	25	-115
8	5	375	50%	-349	300	-500	-75	-151
9			60%	-435		-616		-182
10	i* value	25.41%			23.57%		16.89%	

	A	B	C	D	E	F	G	H
1		Filter 1		Filter 1	Filter 2			
2	Year	Cash flow, $100	Rate, i%	PW, $ 100	Cash flow, $100	PW, $100	Incremental cash flow, $100	Incremental PW, $100
3	0	-1,000	0%	875	-1,500	1,000	-500	125
4	1	375	10%	422	700	467	325	46
5	2	375	20%	121	600	103	225	-19
6	3	375	30%	-87	500	-158	125	-71
7	4	375	40%	-237	400	-352	25	-115
8	5	375	50%	-349	300	-500	-75	-151
9			60%	-435		-616		-182
10	i* value	25.41%			23.57%		16.89%	

- The cash flow sign tests for each filter indicate no multiple rates.
- The incremental cash flow sign test does not indicate the presence of a unique positive root; however, the second rate is an extremely large and useless value.



- The PW values of filter 1 and filter 2 cash flows are plotted for *i* values ranging from 0% to 60%.
- Since the PW curves cross each other at approximately $i^* = 17\%$, the rate does exceed the MARR of 15%.
- The higher-cost filter 2 (Purely Heaven) is selected.



- The PW curve for the incremental cash flows series is plotted.
- As expected, the curve crosses the PW = 0 line at approximately 17%, indicating the same economic conclusion of filter 2.

- The incremental breakeven rate is 16.89%, and the MARR is 15%.
- The MARR is lower than breakeven; therefore, the *incremental ROR analysis* results in correctly selecting filter 2.
- But if only the i^* values were used, filter 1 would be wrongly chosen because its i^* exceeds that of filter 2 (25.41% > 23.57%).
- This error occurs because the rate of return method assumes reinvestment at the alternative's ROR value, while PW and AW analyses use the MARR as the reinvestment rate.
- This is sometimes called the **ranking inconsistency problem** of the ROR method. *The inconsistency occurs when the MARR is set less than the breakeven rate between two revenue alternatives*.
- <u>The conclusion is simple:</u>
- If the ROR method is used to evaluate two or more alternatives, use the **incremental cash flows and** Δi^* to make the decision between alternatives.